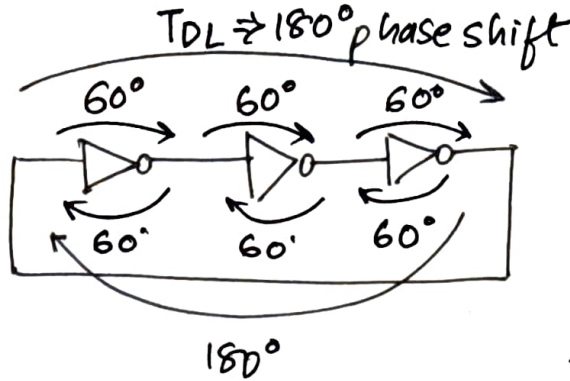




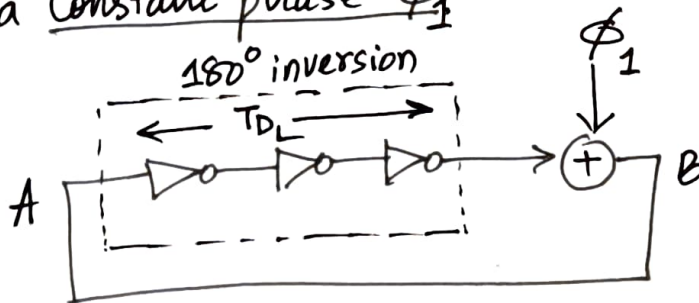
$\therefore \text{total} = 360^\circ \checkmark$

$T_{DL} = T_0/2$ complete oscillation
only one way.

$T_0 = 6 T_{DL}$ gate delay.

$3T_0$

- Add a Constant phase ϕ_1



frequency adjustment

$2\pi \frac{T_{DL}}{T_0} + \phi_1 = \pi$

new T_{osc}

* Total phase shift has to be 180°

"fosc"

NOTE: Oscillation frequency shifts in response to phase shift changes.

In case of 3 inverters : $T_{DL} = 180^\circ$ $T_0 = 360^\circ$

$\therefore 2\pi \left(\frac{180}{360} \right) + \phi_1 = \pi$

$\pi + \phi_1 = \pi$

$\phi_1 = 0$ (phase shift)

NOTE 2 (1) Gate delays don't change

(2) T_{osc} and f_{osc} changes.

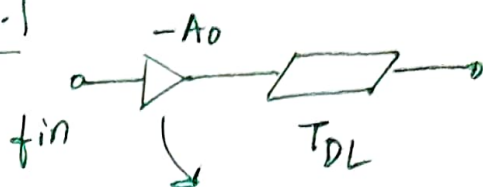
$\rightarrow 2\pi \left(\frac{T_{DL}}{T_{osc}} \right) + \phi_1 = \pi$

$\cdot 2\pi \left(\frac{f_{osc}}{2f_0} \right) + \phi_1 = \pi$

$\rightarrow T_{DL} = \frac{T_0}{2} = \frac{1}{2f_0}$

$f_{osc} = f_0 - f_0 \frac{\phi_1}{\pi}$

Example 3.1



phase shift as???
a f's of i/p freq

no phase shift; inverting.

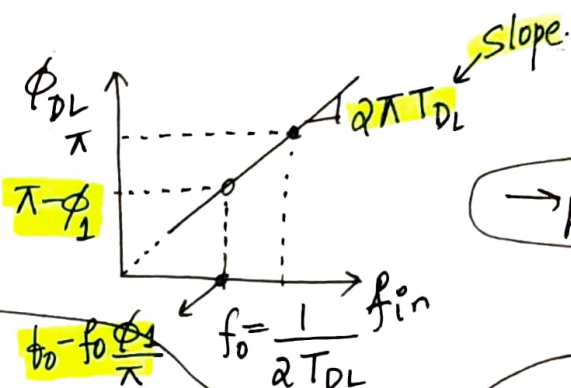
→ $T_{DL} \rightarrow \text{constant}$

→ independent of f_{in} : input freq

ϕ_{DL} (rad) phase shift $\Rightarrow \phi_{DL} = 2\pi \left(\frac{T_{DL}}{T_{in}} \right)$

$$\phi_{DL} = 2\pi T_{DL} \cdot f_{in}$$

$\phi_{DL} \propto f_{in}$



→ phase shift ϕ_{DL} crosses π

$$\text{at } f_0 = \frac{1}{2T_{DL}}$$

→ if loop?



ϕ_1 : positive phase shift

$\checkmark -A_0$: inverting: total phase shift = 180°

$$\therefore \phi_{DL} = \pi \Rightarrow \phi_{DL} = 2\pi T_{DL} f_0 = \pi$$

$$f_0 = \frac{1}{2T_{DL}}$$

* ϕ_1 is inserted: positive

* 180° is required

* delay line provides: $(\pi - \phi_1)$ only

$$\therefore \pi - \phi_1 = 2\pi T_{DL} f$$

$$f = \frac{\pi - \phi_1}{2\pi T_{DL}} = \frac{1}{2T_{DL}} - \frac{\phi_1}{2\pi T_{DL}}$$

Derivation of $f = f_0 - \frac{\phi_1}{2\pi T_{DL}}$

$$f = f_0 - \frac{\phi_1}{2\pi T_{DL}}$$

$$f = f_0 - \frac{f_0 \phi_1}{\pi}$$

* **NOTE**: Phase noise in the delay line causes frequency noise in the oscillator

① if ϕ_1 is phase noise ($\phi_{n,DL}$) of delay line open-loop phase noise then oscillation freq becomes:

$$f = f_0 - \frac{f_0}{\pi} \phi_{n,DL}$$

② oscillator output phase due to Delay line phase noise

$\phi_{n,OSC}(t)$ ~~$\Rightarrow \int 2\pi \frac{f_0}{\pi} \phi_{n,DL}(t) dt$~~

closed-loop phase noise

$\phi_{n,DL}$
↓
decreases, increases.

$\therefore \phi_{n,OSC}(t) = \int 2\pi f_0 \phi_{n,DL}(t) dt$

converts Hz to rad/s

$$\phi_{n,OSC}(t) = \int 2\pi \frac{f_0}{\pi} \phi_{n,DL}(t) dt$$

if $g(t) = \int \omega(t) dt$

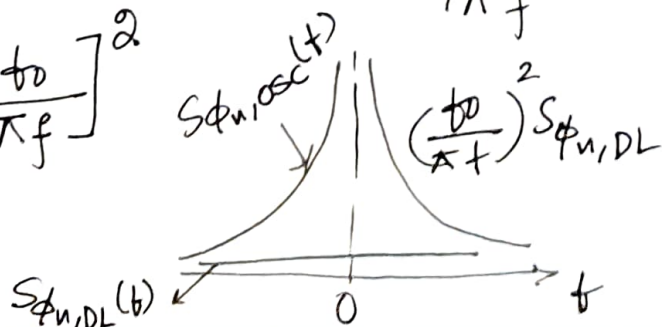
spectrum of $g(t)$ & $\omega(t)$ (chap 2)

$$S_g(t) = \frac{1}{4\pi^2 f^2} S_{\omega}(t)$$

③ Spectral Relationship (phase Noise Shaping)

$$S_g(t) = \frac{1}{4\pi^2 f^2} S_{\omega}(t) \Rightarrow S_{\phi_{n,OSC}}(f) = \frac{4f_0^2}{4\pi^2 f^2} S_{\phi_{n,DL}}(t)$$

$$\Rightarrow S_{\phi_{n,OSC}}(t) = S_{\phi_{n,DL}}(t) \cdot \left[\frac{f_0}{\pi f} \right]^2$$



- * Applies to both Linear & Non-linear circuits.
 $\Rightarrow$ no small-signal operation.

Example 3.2 open-loop & closed-loop phase noise

- * oscillator oscillates at f_0 freq.
- * open-loop ~~freq~~ $\phi_{open}(f_0) = \pi$ (180° for inversion)
 phaseshift
- * if ϕ_1 is added : $\phi_{open} + \phi_1 = \pi$ ✓

① change in oscillation freq, due to addition of ϕ_1 :

$$\Delta f = \frac{\phi_1}{\left(\frac{d\phi_{open}}{df} \right)}$$

↗ phase slope

↘ frequency deviation is proportional to phase noise divided by ^{phase} slope of loop.

② Closed-loop phase noise $\phi_{closed}(t)$?

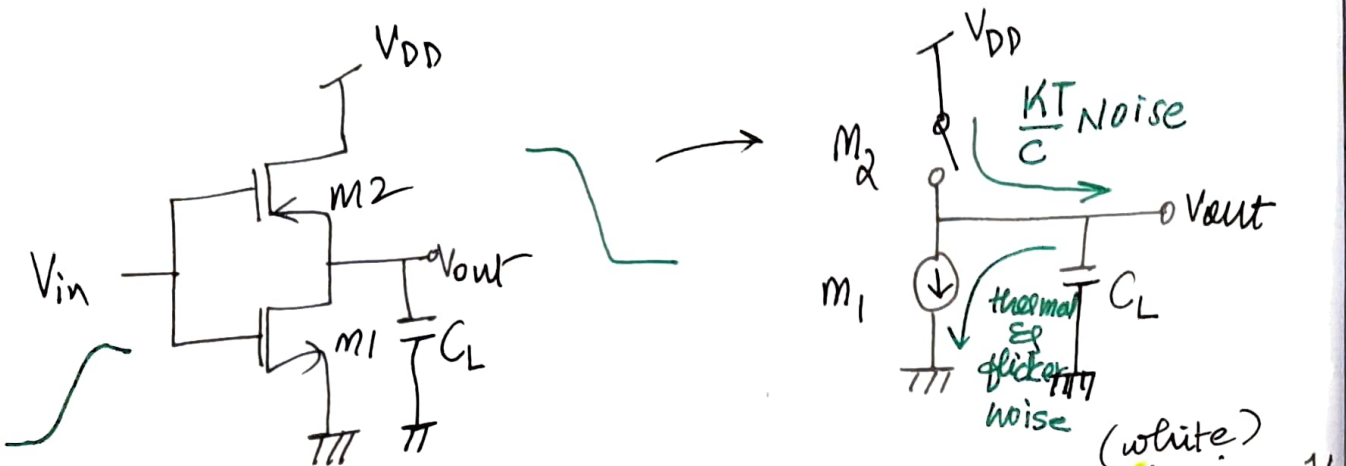
$$\phi_{n,closed}(t) = \int 2\pi \frac{\phi_{n,open}}{d\phi_{open}/df} \cdot dt$$

③ Spectrum

$$S_{\phi_{n,closed}}(f) = \frac{1}{\left| \frac{d\phi_{open}}{df} \right|^2} \cdot \frac{1}{f^2} \cdot S_{\phi_{n,open}}(f)$$

↘ $1/f^2$ -shaping function.

Noise Sources in MOSFET CMOS inverter



- NOTE**
- ① NMOS & PMOS produce $\rightarrow$ thermal noise: $1/f$
white noise: $1/f$
flicker
 - ② Can cause jitter in output
 - ③ noise from NMOS & PMOS is injected to capacitance C_L

Calculate phase noise of Ring Oscillator using 1 CMOS inverter

N-stage ring oscillator:

white Noise: $S_{\phi_n, \text{white}}(f) = \frac{f_0^2}{f^2} \left[\frac{1}{2I_D^2} (S_{I, \text{NMOS}} + S_{I, \text{PMOS}}) \right]$

flicker noise: $S_{\phi_n, \text{flicker}}(f) = \frac{f_0^2}{f^2} \frac{1}{4NI_D^2} (S_{1/f, \text{NMOS}} + S_{1/f, \text{PMOS}})$

$\frac{2KT}{I_D V_{DD}}$

① white noise phase noise is independent of C_L and N : stage

② I_D, V_{DD} , MOSFET parameters matter the most.

③ $S_I = 2KT \gamma g_m$
 $g_m = \frac{2I_D}{(V_{DD} - V_{Th})} \Rightarrow S_{\phi_n, \text{white}}(f) = \frac{f_0^2}{f^2} \left\{ \frac{4KT \gamma}{I_D (V_{DD} - |V_{Th}|)} \right\}$

Remember : You can compute osc phase noise from phase noise of single inverter

Example 3-3 & 3-4

osc freq & phase noise if widths are doubled???

→ ① oscillator freq : remains same
(delay per stage) ↑

② white noise — thermal noise : — HALVED

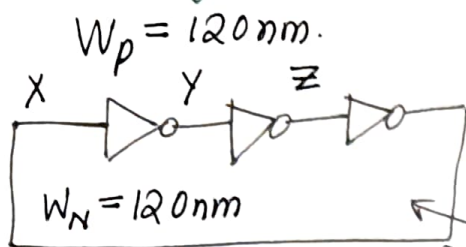
③ flicker noise ↘
↗ I_D doubles
↘ noise current doubles

HALVED : gate area ↑↑↑

④ GREATER WIDTH $\equiv$ Lower Phase Noise

3.2 Preliminary Design Rules

Steps



$$f = 2 \text{ GHz}$$

* start with min-width & min-no. of inverters

* Ex: i) Tech: 40nm

$$\rightarrow L_{\min} = 40 \text{ nm}$$

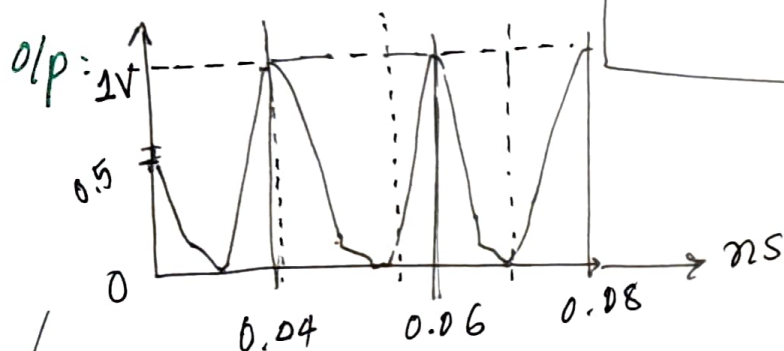
ii) min Width: $W_{\min} = 120 \text{ nm}$
(nmos pmos)

iii) use 3-stage, $\frac{W}{L} = \frac{120}{40}$ n

p & n trans

: First Simulation :

3-stage,
 $V_{DD} = 1 \text{ V}$,
room temp



must try in Cadence.

Current? $\rightarrow$ i) PMOS ON : draws current from $V_{DD} \rightarrow$ charges C_L
ii) I_{DD} peaks at Rising Edge $\rightarrow 3 \times f_{osc} = I_{DD}$

Results. $\frac{1}{T_D}$

* Osci freq = 45 GHz $\Rightarrow$ Gate Delay $T_D = 3.7 \text{ ps}$

① interconnect capacitance

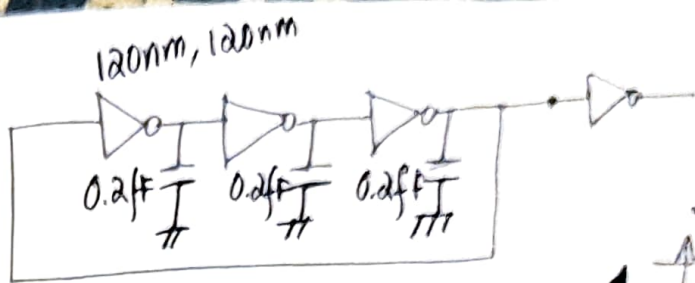
② Input capacitance of loads

other factors

V. important

③ Worst Case : eg., low V_{DD} , high temp
Scenario

Slow, slow process corner.

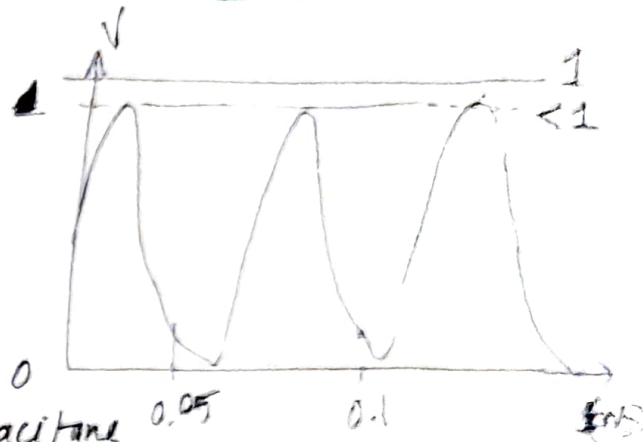


$$V_{DD} = 0.95V$$

$$T = 75^{\circ}C$$

SS corner

Worst Case Scenario

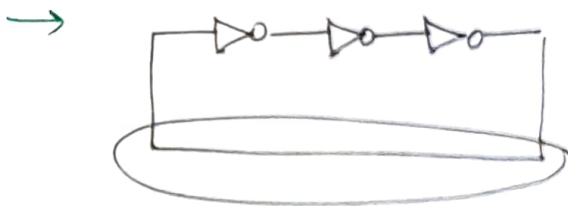


0.2fF $\rightarrow$ interconnect capacitance
 o/p is buffered with a fanout

Result : 21GHz osc freq $\rightarrow$ slow transistors
 $\rightarrow$ heavy cap.

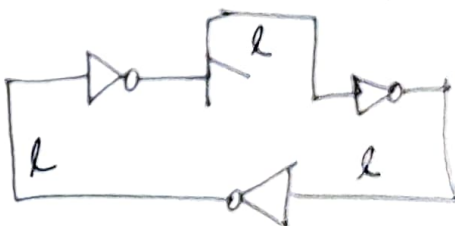
Example 3.6

Layout Optimization



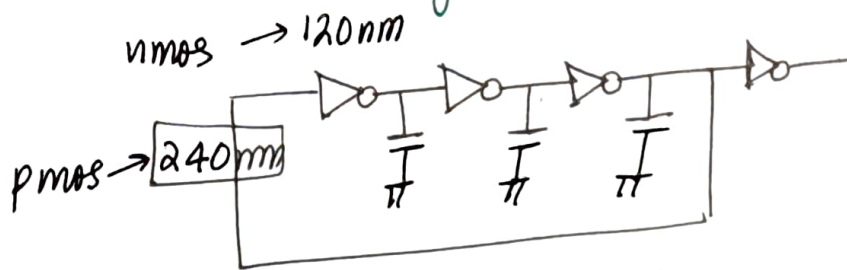
long wire $\rightarrow$ increases capacitance
 $\rightarrow$ unequal Delays.

Soln:



same lengths

Improve Swing : stronger PMOS.



PMOS : 240nm width
NMOS : 120nm width

$V_{DD} = 0.95V$
Temp = 75°C
SS corner

Result → 22.4GHz → incr Osc freq

✓ PMOS pull up faster
✓ outweighs effect of capac

Current : 60μA
Avg

power consumed : 57μW

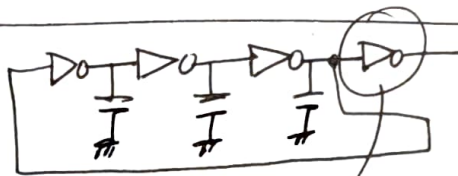
[Supply sensitivity] ??

$$P = 3 f_0 C_{tot} V_{DD}^2$$

K_{VDD} → change V_{DD} by 10mV → $\Delta f_{osc} = 50.2MHz$

$$\therefore K_{VDD} = 50.2GHz/V$$

practical



why 4th inverter?????

- Buffer : 4th inverter acts as a load → mixer
→ counters
→ block of PLL
→ Etc....

3.3) Obtaining Desired frequency

previously..... obtained = 22.6 GHz desired = 2 GHz
freq freq

- 4 options
- ① Add Capacitance to Each node
→ slow down transistors.
 - ② increase no. of stages
 - ③ increase length of Each transistors
 - ④ Divide o/p frequency by a factor of 10 or 11

3.3.1 Greater Node Capacitances.

$$\text{osc freq} \propto \frac{1}{\text{total Cap at Each Node}}$$

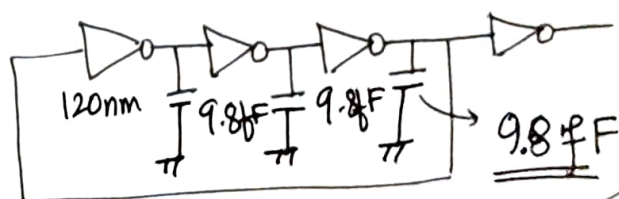
Simulation : $C_{\text{tot}} = 0.93 \text{ fF}$

* Reduce 22.6 GHz $\xrightarrow{\text{to}}$ 2 GHz

$$C_{\text{tot, new}} = \left(\frac{22.6}{2} \right) \times C_{\text{tot}} = 10.5 \text{ fF}$$

* add $10.5 \text{ fF} - 0.93 \text{ fF} = 9.6 \text{ fF}$ to Each node

240nm



Result : 1.93 GHz

rise & fall times
increase
gate delay $T_D \uparrow \uparrow$

K_{VDD}
4.3 GHz/V

power dissip $\Rightarrow P = 3 f_0 C_{\text{tot}} V_{DD}^2 \rightarrow \text{SAME}$

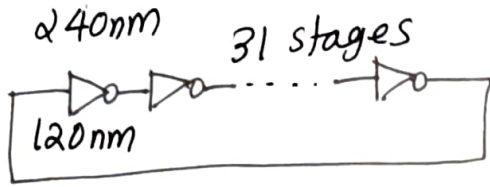
$\frac{K_{VDD}}{f_0} = \text{SAME} = \text{changes only if invt change}$

$f_0 \downarrow C_{\text{tot}} \uparrow$

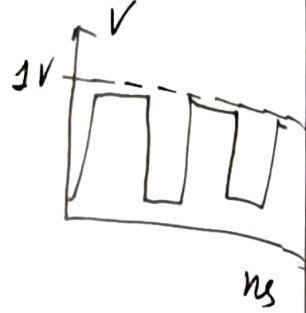
[3.3.2] Greater No. of Stages: N

→ increase in No. of Inverters $\uparrow\uparrow\uparrow$
 ⇒ increase in overall delay $\Rightarrow f_{osc} \downarrow\downarrow$

Ex: 31 stages



Same ref design



Result → 2.4 GHz
 → unity fanout: SAME
 → sharp edges → Each invt operates at same speed

Power? : $P = N f_0 C_{tot} V_{DD}^2$

but: $\frac{P}{f_0} = \frac{1}{2NT_D} \Rightarrow P = \cancel{N} C_{tot} V_{DD}^2 \cdot \frac{1}{2\cancel{N}T_D}$

$P = \frac{C_{tot} V_{DD}^2}{2T_D} \rightarrow$ independent of (N) .

3 & 31 stages

Use the SAME amt of POWER.

K_{VDD} : 4.9 GHz/V

$\frac{K_{VDD}}{f_0} = \text{SAME}$

3.3.3 Greater transistor lengths: L

increase length:

① investors become WEAK

- less current drive
- switching is slow

② Input Capacitance risks

- Gate area increases
- $Cap_{\text{ground}}^{\text{to}}$ increases

③

flicker noise decreases

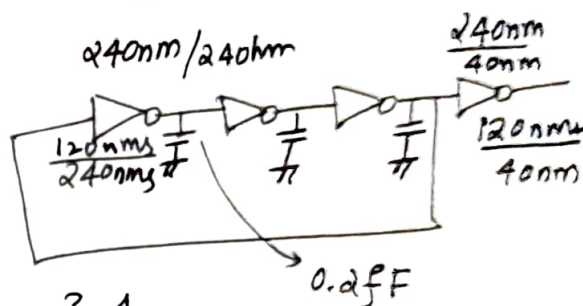
- larger channels have less Low-freq ($1/f$) noise

To Reduce 22.6GHz to 2GHz ???

$$\text{Delay} \propto L^2$$

drive capacitance $\uparrow$

$$m = \sqrt{\frac{22.6\text{GHz}}{2\text{GHz}}} = \sqrt{11.3} = 3.4$$



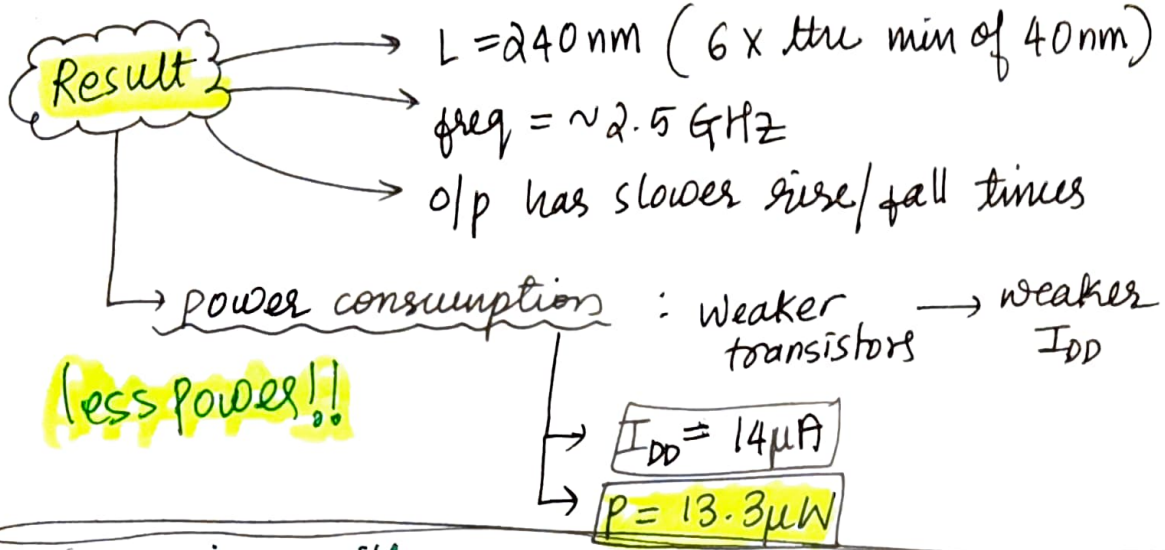
$\Rightarrow$ Lengths have to be increased by $\sqrt{11.3} = 3.4$ times to reduce the frequency factor of ~ 11 (22.6GHz to 2GHz)

BUT: other factors: parasitic & drain capacitances

$\hookrightarrow$ so increase L little more than $3.4 \times$

~~for every 'm' increase in L ; $\sqrt{m}$ times~~

$\hookrightarrow$ L increased by a factor of (m) , gate delay goes up by (m^2)



Comparing with 1.93 GHz model.

- Average current : $60 \mu\text{A}$ to $14 \mu\text{A}$ drop
- Power : $57 \mu\text{W}$ to $13.3 \mu\text{W}$ drop.

* inverters are weakened by a factor of 6.

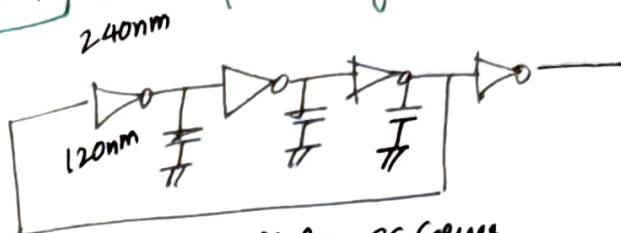
$$\begin{aligned} &\rightarrow 2.5 \text{ GHz vs } 1.93 \text{ GHz} \\ &\rightarrow \text{power reduction factor} = 6 \times \left(\frac{1.93}{2.5} \right) = 4.6 \\ \therefore &\rightarrow I_{DD} = \frac{60 \mu\text{A}}{4.6} = 13 \mu\text{A} \rightarrow \text{very close to simulated } 14 \mu\text{A} \end{aligned}$$

$$K_{VDD} = 4.96 \text{ GHz/V}$$

NOTE

Phase noise increases as power reduces.

3.3.4 Frequency division



$V_{DD} = 0.95V$; $T = 75^\circ C$, SS corner

o/p of high freq ring osc is fed to freq divider counter.

* counter consumes more power

* no change in (K_{vco}/f_0)

Summary

- ① Adding Cap: transitions slow; waveform not as sharp
- ② Stages: Edges sharp; area $\uparrow\uparrow$; complexity $\uparrow\uparrow$
- ③ Length: Lower paras $\rightarrow$ increases rise/fall times $\rightarrow$ phase noise $\uparrow\uparrow\uparrow$
- ④ freq divide: more power consumption $\rightarrow$ no improvement seen.

3.4 Phase Noise Consideration

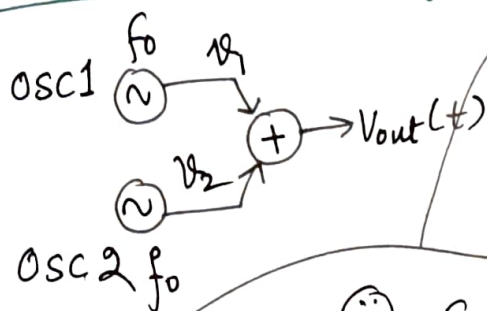
* phase noise simulations use: $V_{DD} = 0.95V$; SS Model
 $T = 750C$ HERE.

* more power = less phase noise

$$\text{power} \propto \frac{1}{\text{phase noise}}$$

phase noise relates to osc freq

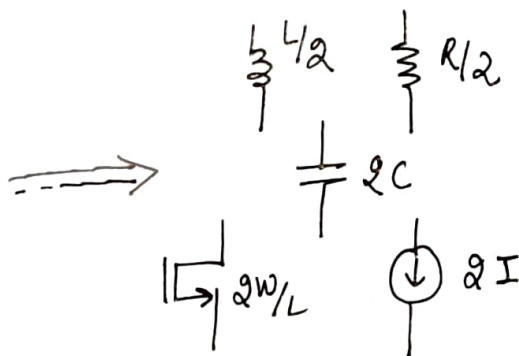
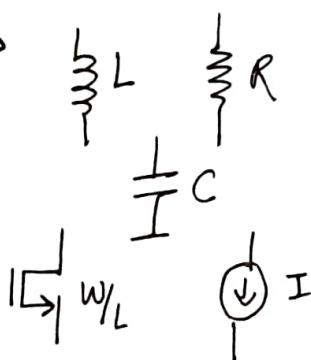
Example 3.7 Reducing phase noise by 3dB



(i) adding identical o/p's of oscillators
 → combined phase noise is Halved.

(ii) Scale up parameters.

- Voltage swing & freq SAME
- phase noise is HALF



3.4.1 Transistor Noise Simulation

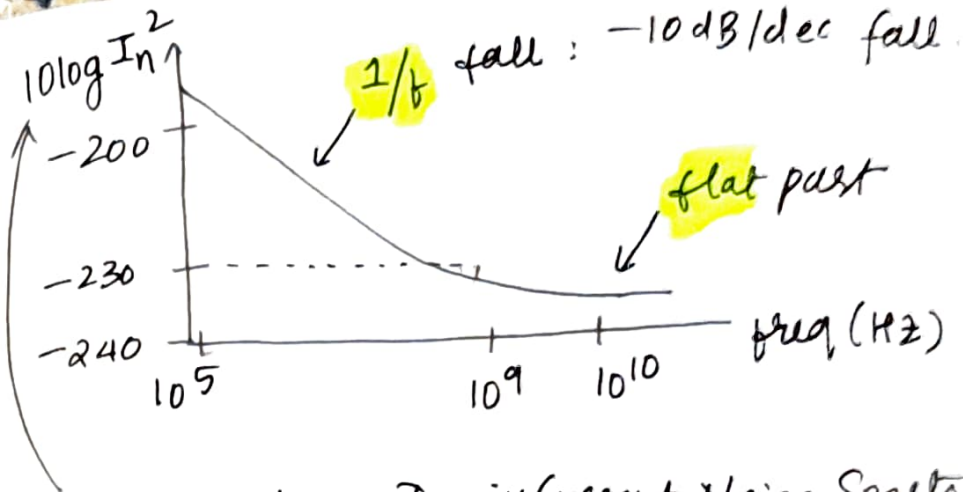
- Transistor noise contributes in phase noise
 - thermal noise (flat across freq)
 - flicker (dom at low-freq)

- Simulations for 120nm/40nm NMOS:

① flicker noise drops at $-10dB/dec$ from 100KHz to 10sMHz

② higher freq ($\sim 500MHz+$) → Noise flattens out.

→ flat part is thermal noise floor



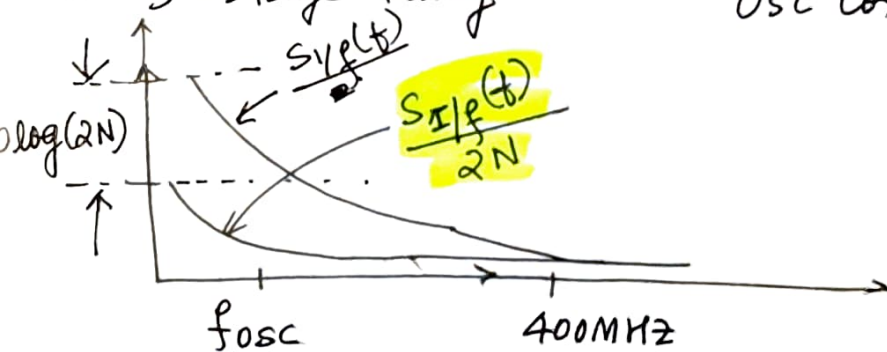
thermal noise $\rightarrow -234 \text{ dB}$: matches the graph.

$S_{\text{thermal}} = 4kT\gamma g_m$

Example 3.8 Flicker noise corner in Ring Osc.

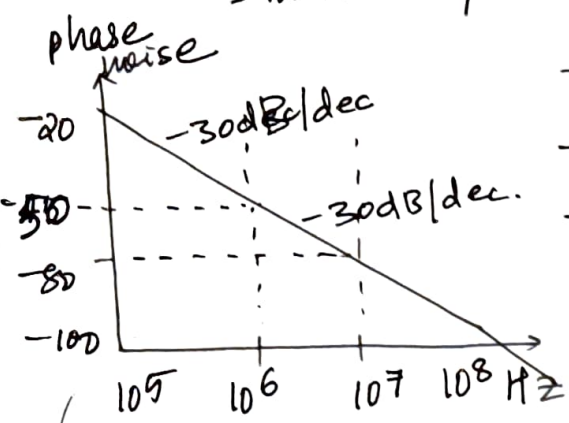
* Oscillator's flicker noise corner is LOWER than of a single transistor by a factor $(2N)$
 $N \rightarrow \text{stages}$

$\therefore$ 3-stage Ring Osc $\rightarrow$ Osc corner $\approx \frac{500 \text{ MHz}}{6} \approx \underline{\underline{83 \text{ MHz}}}$



3.4.2 Reference Osc Phase Noise [Fig 3.12(a)]

- simulated phase noise for 22.6 GHz ref ring



- drops by 30dB/dec
- corner freq ≈ 90 MHz
- Very high phase noise = -47dBc at 1MHz offset
- why?
- we need to normalise by f_{osc} , f , power consumption.

Example 3.9 Oscillator FOM figure of Merit

$$FOM = \frac{1}{(\text{phase noise}) \times (\text{power})} \times \left(\frac{f_0}{f} \right)^2$$

- * higher FOM → Better oscillation
- * phase noise ↓ in high freq. ∴ FOM improves from low to high freq.

Theoretical vs Simulated phase Noise

- use measured current noise I_n to cal theoretical phase noise
- ✓ match with simulation
- at 100MHz offset: predicted ≈ -104 dBc/Hz
- at 1MHz: p: -46 dBc/Hz ✓

look into diagrams while simulating

Example 3.10 FOM Calculation

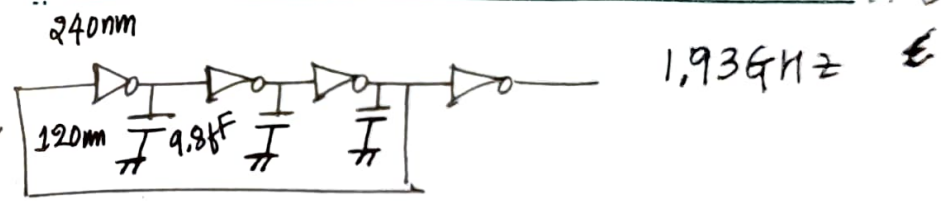
$$FOM = \frac{1}{\text{phas } N \times \text{pavg}} \times \left(\frac{t_0}{t} \right)^2$$

Example 3.11 Lowering Phase Noise

• Want : -120 dBc/Hz at 100 MHz f [16 dB better than referenc osc]
Same f_{osc}

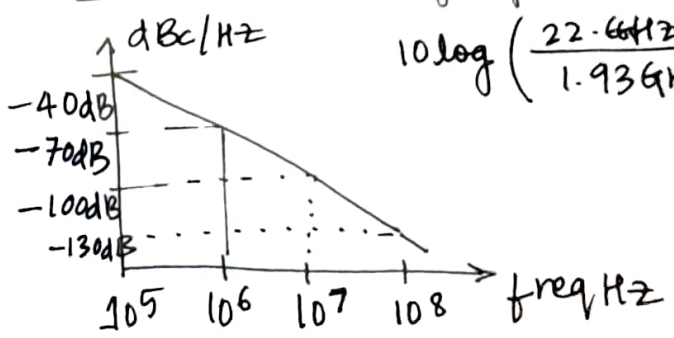
• Solu : $\rightarrow$ phase noise red by 16 dB $\rightarrow$ Increase power consumption by a factor $10^{1.6} \approx 40$
 $\rightarrow$ Scale widths / others by 40.
 $\rightarrow$ power rises by : $40 \times 60 \mu\text{A} \times 0.95 \text{ V} = 2.3 \text{ mW}$
 PN reduces by 16 dB

3.4-3 First 2GHz Osc Phase Noise - Adding Cap



$\rightarrow$ Keep power the same ; lower freq by adding Capacitances.

$\rightarrow$ Phase Noise graph



$$10 \log \left(\frac{22.6 \text{ GHz}}{1.93 \text{ GHz}} \right) = 21.5 \text{ dB} \leftarrow \text{shld fall by.}$$

Conclusion

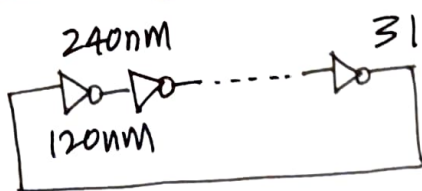
* for a given Osc freq, prefferable to increase no. of stages than to adding/incre capacitance.

• FOM doesnt change
 • power & PN scale together

3.4.4

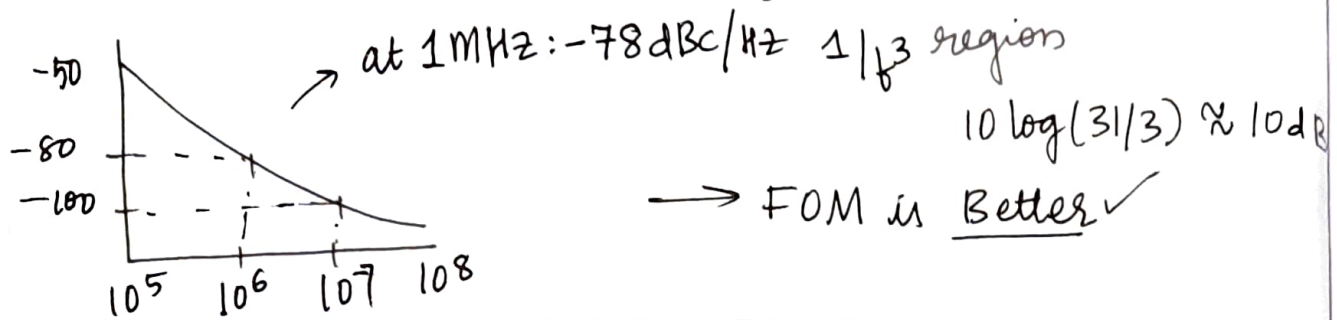
Second 2GHz Osc Phase Noise

MORE stages!



$$\rightarrow f_0 = 2.3 \text{ GHz}$$

$\rightarrow$ Phase noise decreases in



3.4.5

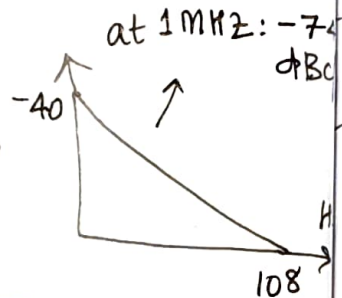
Third 2GHz Osc PN

Longer lengths

$\rightarrow$ longer lengths of NMOS & PMOS

$\rightarrow$ Less power: $P = 13.3 \mu\text{W}$ (go back to check)
 $\rightarrow$ due to less node cap.

$f = 2.5 \text{ GHz}$ ring phase noise graph



Result $\rightarrow$ PN slightly higher
 $\rightarrow$ **FOM is the highest**
 $\rightarrow$ Best phase - to - power ratio

Example 3.12 Comparison b/w 3.4.4 & 3.4.5

⊛ Compare $1/f^3$ regions

$\rightarrow$ at 100MHz: phase noise is

Conclusion: increase length $\rightarrow$ Best FOM.

3.4.6 4th 2-GHz Osc Phase Noise

* Use a fast Oscillator plus a freq divider.

* ~~Input~~ o/p freq is divided by M.

$$V_{div}(t) = V_0 \cos \left[\frac{\omega_0 t + \phi_n(t)}{M} \right]$$

∴ phase is also divided by M !!

* o/p phase noise is lower by a factor of M

$20 \log_{10}(M)$ in dB

OSC freq is divided by M $\Rightarrow$ phase is divided by M $\Rightarrow$ phase noise is divided by M factor
 No change in FOM, jitter. $\because$ OSC freq is reduced. $\Leftarrow$ it is $20 \log(M)$ in dB $\Leftarrow$ is divided by M factor

Example 3.13 Improve Third 2-GHz Osc's ~~Second~~ phase noise

It is -74 dBc/KHz

Power: rises to 8.2 mW

wanted: -100 dBc/KHz (lower phase noise)

$\Rightarrow$ increase power by a factor $\sim 631 \rightarrow 10^{2.8} \approx 631$

$\Rightarrow$ increase Widths of NMOS & PMOS: $(76 \mu\text{m}/240\text{nm})_{\text{NMOS}}$
 $(152 \mu\text{m}/240\text{nm})_{\text{PMOS}}$

Summary

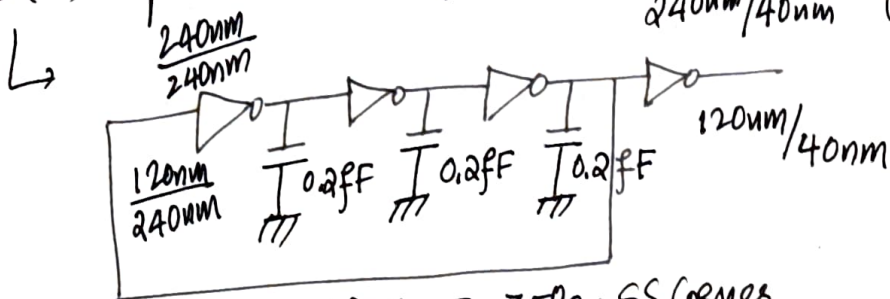
1. Best: Increase length: highest FOM & Phase noise/watt
2. Next Best: Increase No. of stages: sharpest Edges; improved flicker noise
3. Adding Cap: Reduces Osc freq: No change in FOM
4. Freq Division: Reduces Phase Noise: Not absolute jitter, No change FOM.

* for Spec. ref to pg: 83 textbook

3.5

FREQUENCY TUNING

3.16(a) optimised Ring oscillator



$V_{DD} = 0.95V$; $T = 75^{\circ}C$; SS corner

develop methods for tuning frequency of Inverter-Based Ring Osc. VCOs

(3.5-1) Tuning Considerations - 5 aspects

① Tuning freq Range

- wide enough
- able to cover all required frequencies,
- all PVT variations & application needs

② VCO Gain (K_{VCO})

→ Low K_{VCO} is desired

→ K_{VCO} (Hz/V): freq changes per volt of $V_{control}$.

- High K_{VCO} makes VCO more sensitive to noise on control line, causing large freq & phase fluctuations.

→ Larger Range gives Lower K_{VCO} .

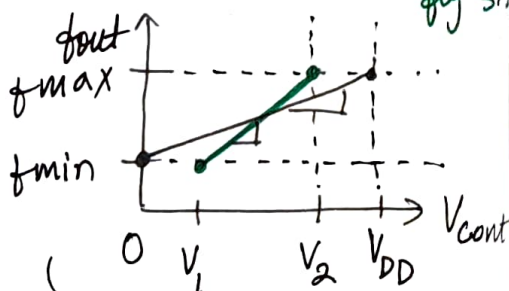
Rule of thumb

→ Keep $K_{VCO} < 0.1 f_m$ Hz/V

where $f_m = \left(\frac{f_{max} + f_{min}}{2} \right)$

Ex: $f_m = 2.5GHz$

→ $K_{VCO} < 250MHz/V$



allowable range

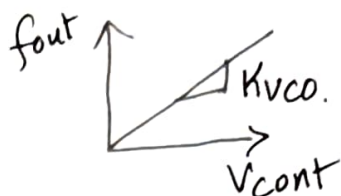
$[V_1, V_2] \rightarrow$ higher K_{VCO}

$[0, V_{DD}] \rightarrow$ lower K_{VCO}

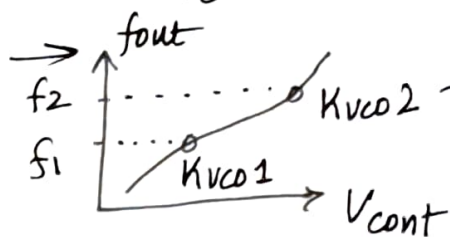
fig 3.1

③ Linearity of tuning

how much the slope $[K_{VCO}]$ varies



K_{VCO} shld vary by no more than 20% across range



- f^y changes from f_1 to f_2
- Expect K_{VCO2} NOT to change more than 20% than K_{VCO1} .

④ Constant Internal Voltage Swing

→ osc signal swing constant across tuning range

- Why?? ① Amp changes leak into phase noise
- ② AM-to-PM conversion is minimised
- ③ Power, duty cycle, K_{VCO} stay in check

check into pseudo-diff vco's.

⑤ Stable Phase Noise Across Range

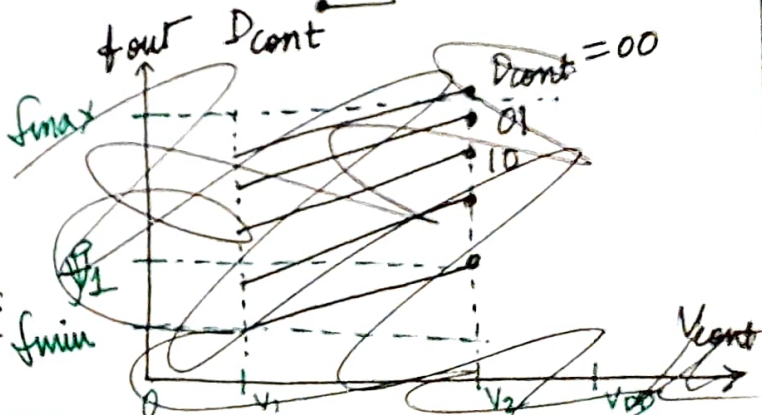
PN shldnt change significantly as freq tuned.

3.5.2 Continuous & Discrete Tuning

- Larger freq range make VCOs sensitive to noise on its control line.

→ digital - discrete tuning
→ Analog - continuous.

- choose a path controlled by V_{cont} weak enough
to hence K_{VCO} Low enough
also providing discrete jumps in freq using dig control.



• Continuous tuning : Varies frequency with an analog control voltage (V_{cont}) .. but limited range to keep K_{vco} low

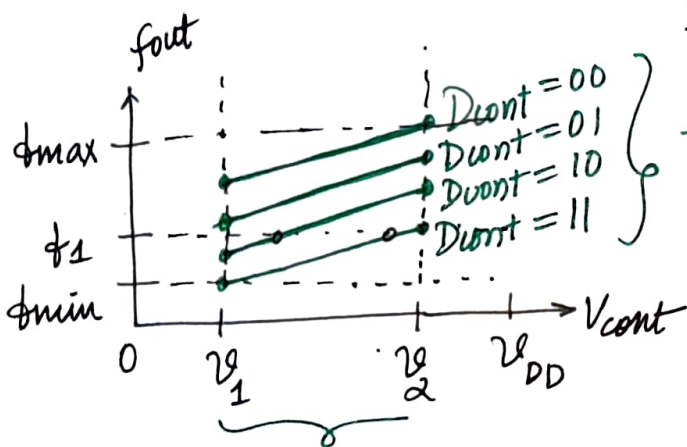
• Discrete tuning : $\rightarrow$ selects b/w ^{diff} freq bands
 $\rightarrow$ by switching cap/resistor on/off.

To gether : $\rightarrow$ Analog tuning covers a SMALL freq range for each dig setting

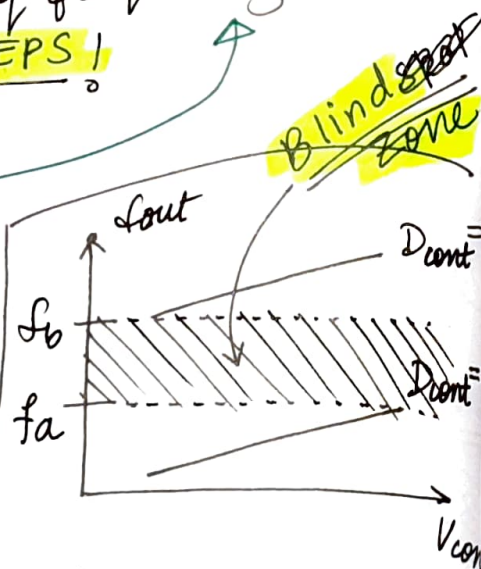
$\rightarrow$ Digital shifts the whole analog tuning curve into high/low

$\hookrightarrow$ full req freq range into

STEPS!



Analog tuning creates a SMALL range



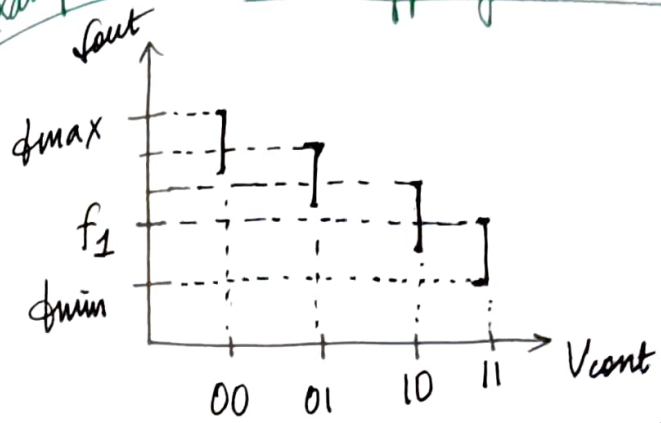
$\rightarrow$ If two adjacent curves do not overlap, VCO fails to cover a certain range $[f_a, f_b]$

✓ Low Sensitivity to Noise

Eg

Wide tuning Range !

Example 3.15 Overlapping bands

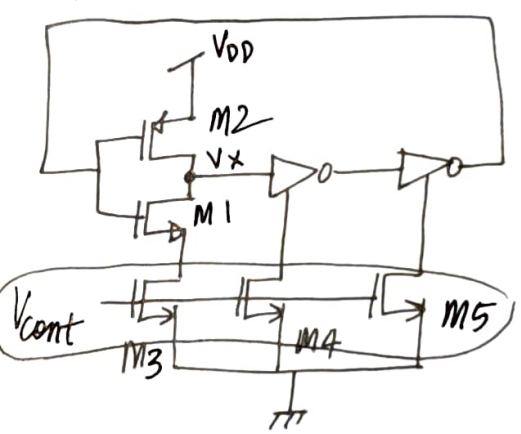


- Rule of thumb :**
- set V_{CO} to cover $\pm 10\%$ to $\pm 20\%$ of center freq
 - choose digital (no. of steps) to cover all freq & still give overlap
 - increase steps if KV_{CO} high

3.5.3 Tuning By Variable Resistance

gate delay depends on $\rightarrow$ (strength of invt) Now $\downarrow$
 $\rightarrow$ capacitances.

* NMOS-Based Variable Resistance (fig 3.30)



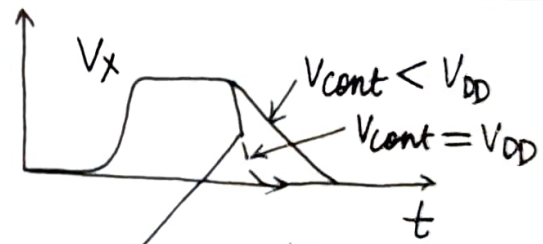
(M3 - M5) $\rightarrow$ Voltage-dep resistors

- Resistance changes with V_{cont}
- if V_{cont} starts at $V_{DD} \rightarrow 0$
- pull-down strength of invt also decreases with it

M3, M4, M5

• triode region when $V_{cont} \approx 0$

• $V_{cont} = V_{DD} \therefore$ NMOS short circuit
 Zero resistance
 high freq

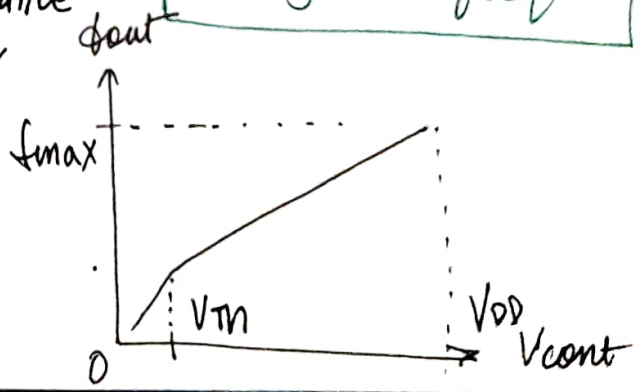


$\rightarrow$ On-resistance increases as $V_{cont} : V_{DD} \rightarrow 0$

Gate delay $\uparrow$ Osc freq $\downarrow$

$V_{cont} = [V_{DD} \text{ to } 0]$

- ✓ resistance $\uparrow \uparrow$
- ✓ inverter weaker.



Problems

i) Asymmetric Transitions

- falling edge slows down
- rising edge SAME: no change
- higher phase noise on one edge

ii) Nonlinear tuning

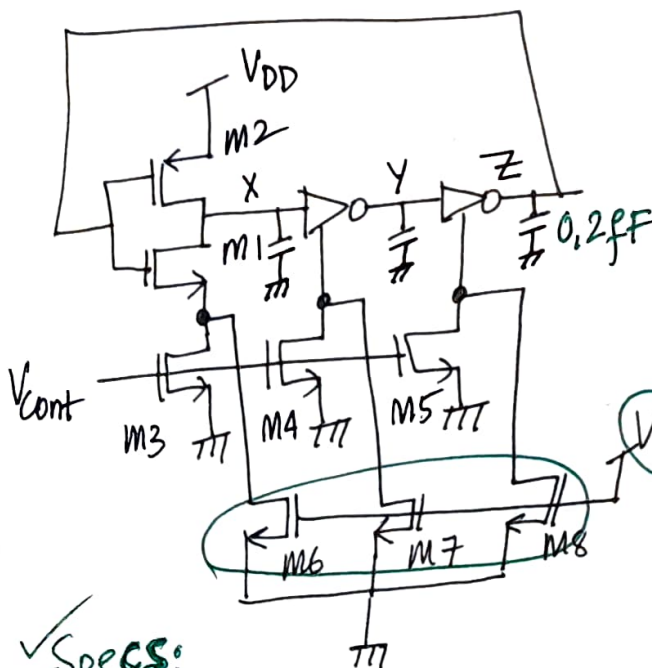
- When V_{cont} very near 0; resistance increases sharply
- causes "sharp drop" in freq
- at $V_{cont} \approx 0$ → no oscillation!

How to Solve No Oscillation??

→ Applying to third 2-GHz Os

▲ Solved

$M_6 - M_8$ → added in parallel
→ gives a min freq at $V_{cont} = 0$



• fixes linearity

M_6 to M_8 : weak NMOS (longer length)

$$f_{max} - f_{min} = 400 \text{ to } 500 \text{ MHz}$$

(10% to 20% of f_m)

• M_3 to M_5 are kept strong — for high MAX freq

• Result → Max freq drops: 2.5 GHz to 2.4 GHz
→ phase noise improves by 3 dB

? $M_6 - M_8$: degenerate the NMOS invt.
: longer channel

✓ Specs:

$$(W/L)_{6-8} = \frac{120 \text{ nm}}{180 \text{ nm}} \text{ (weak)}$$

$$(W/L)_{3-5} = \frac{240 \text{ nm}}{40 \text{ nm}} \text{ (strong)}$$

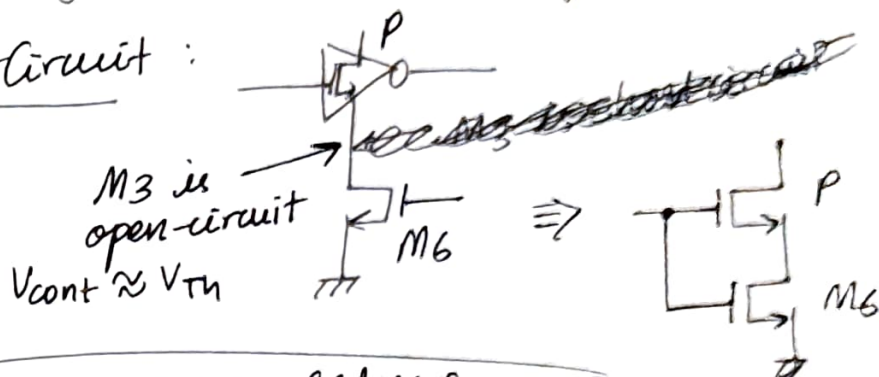
$$(W/L)_N = \frac{120 \text{ nm}}{240 \text{ nm}}$$

$$(W/L)_P = \frac{240 \text{ nm}}{240 \text{ nm}}$$

M_6 to $M_8 \rightarrow V_{cont} \approx V_{Th} \rightarrow$ triode region

- act as "degenerate" NMOS devices in inverter
- making NMOS transistor longer (in inverst)

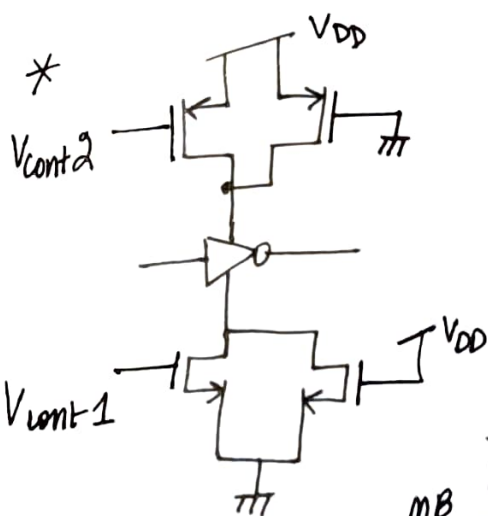
$\therefore$ Equivalent Circuit :



lower osc freq reduces phase noise.

Example 3.16 Symmetric tuning..

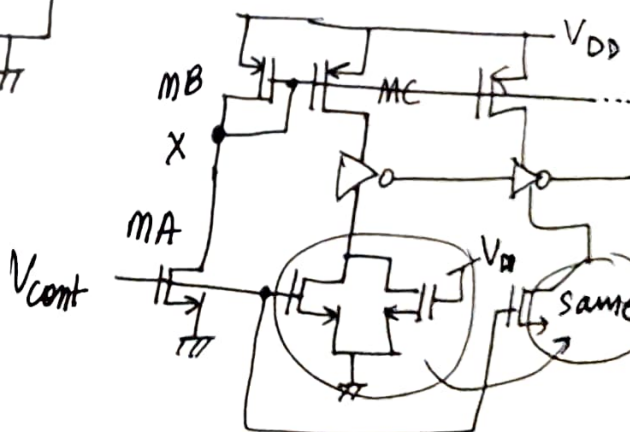
→ tune NMOS & PMOS strength for sym rise & fall



• increase V_{cont1} to weaken NMOS
 decrease V_{cont2} to weaken PMOS

• get $V_{cont1} \rightarrow$ invert it to get V_{cont2}

Current Mirror!



$M_A \gg M_B$
 ensures proper switching

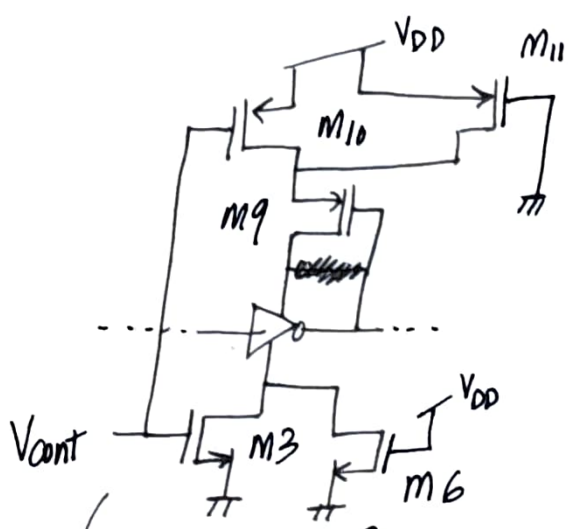
* Added Noise

inverter with M_A & M_B
 add $1/p$ noise

→ corrupt V_{cont2}
 → corrupts Osc freq.

Tuning By Positive Feedback

improve PN



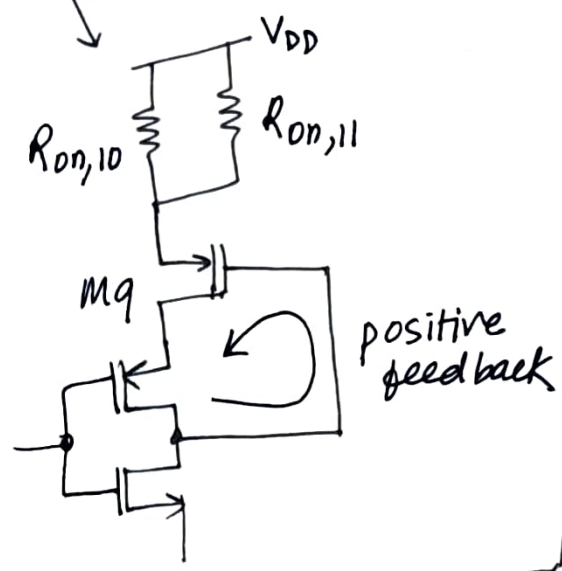
→ M9: PMOS as a feedback path
→ controlled by V_{cont} through M_{10}

* As V_{cont} drops ;
→ turns on more
→ M_{10} 's resistance falls,
→ M9 feedback strengthens
→ Slows down oscillator
→ parallel resistor M_{11}
ensures operation at all values of V_{cont}

Issues → ① M9 must have small dimensions
→ negligible load to inverter
→ ② M_{10} must be weak → to lower freq. by moderate amt.

Results (with linear scaling)

page no: (89).
Reduce K_{vco} & K_{vco} variation?



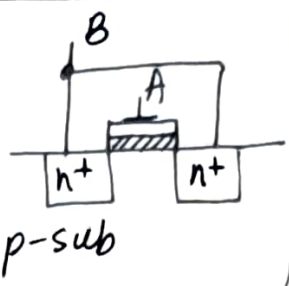
Example 3.17 K_{VCO} Variation

* weaken NMOS (M3) but make Inv't NMOS stronger to compensate
* It keeps max freq
* Result → K_{vco} drops to $\sim 800 \text{ MHz/V}$
→ phase noise increase slightly

3.5.4

Tuning By Variable Capacitance

* Varactors: Voltage-dep Capacitors



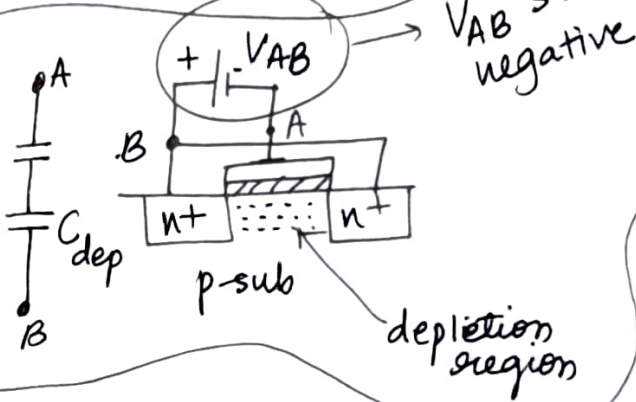
MOS-varactor

→ if V_{AB} is negative;

- e^- are repelled
- form a depletion region

$$C_{AB} = \frac{C_G C_{dep}}{C_G + C_{dep}}$$

series combination
comb of gate oxide
& depletion region



V_{AB} still negative

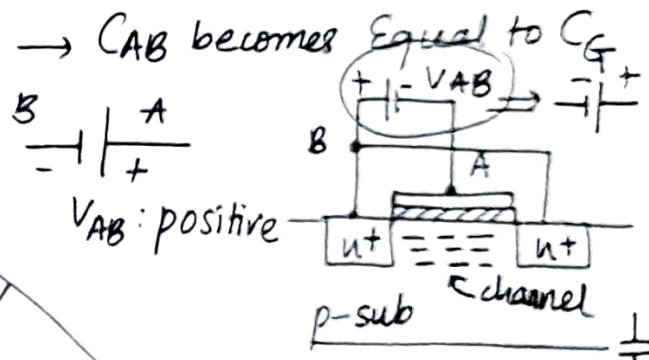
→ if $|V_{AB}|$ decreases

- depletion region gets reduced
- Capacitance increases higher C_{AB}

→ if V_{AB} is sufficiently positive

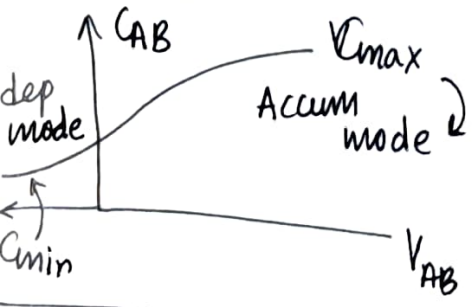
• Accumulation Mode!

↳ channel of e^- forms under oxide



V_{AB} : positive

C-V graph



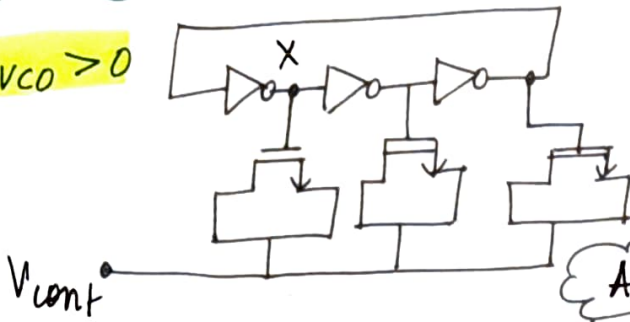
Modelling Varactor

$$C_{AB}(V_{AB}) = \frac{C_{max} - C_{min}}{2} \tanh\left(\frac{q + V_{AB}}{V_0}\right) + \frac{C_{max} + C_{min}}{2}$$

→ for larger $|V_{AB}|$ C_{AB} saturates at C_{min}/C_{max}

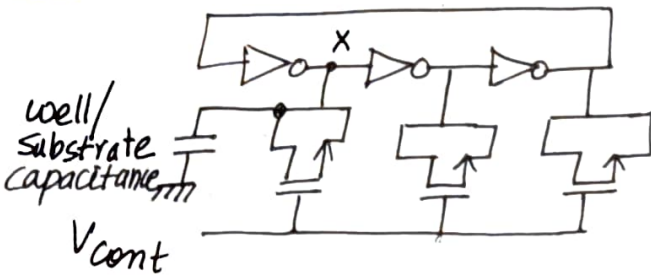
Integrating Variables in Ring Oscillators

(a) $K_{VCO} > 0$



Tuning Range
→ set by Varactor
size

⑥ $K_{VCO} < 0$



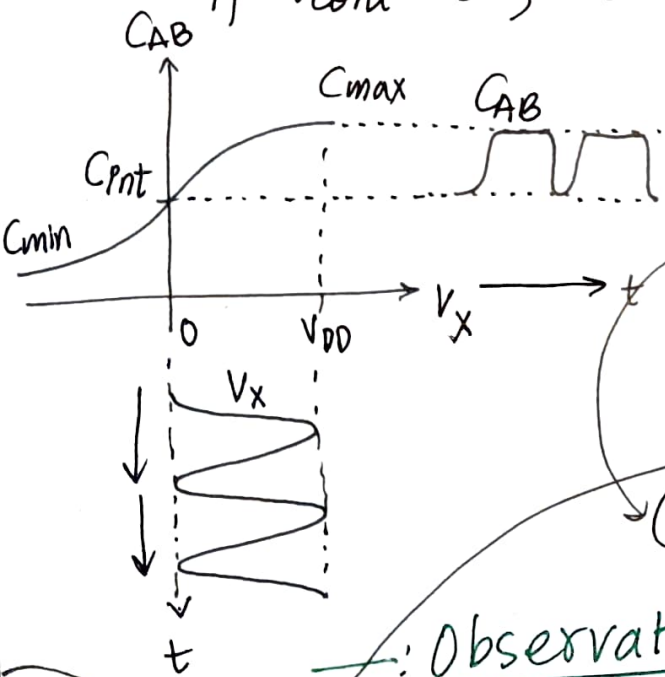
Advantage

↳ Very low noise compared to transistor tuning / variable resist

needs
account variables

Example 3.18 : Behaviour in Operation.

→ voltage across varactors in (a)
if $V_{\text{cont}} = 0$; how does cap vary with time?



- V_x : Gate Voltage of Varactor
Swings from $[0 \text{ to } V_{DD}]$ ← due to V_{DD}
- C_{AB} goes from $[C_{int} \text{ to } C_{max}]$
- $C_{min} \text{ to } C_{int}$: unutilised
 $C_{AB} \neq V_x \neq \text{negative}$

(used the formula)

Observations : -

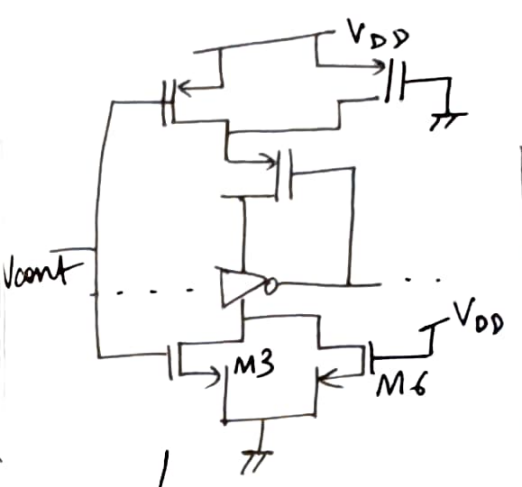
- ① V_{cont} rises from 0 to $V_{DD} \Rightarrow f_o$ also rises $\Rightarrow K_{VCO} > 0$
- ② if gates are connected to $V_{cont} \Rightarrow K_{VCO} < 0$
- ③ for $V_{cont} = V_{DD}$; $\rightarrow$ osc freq drops $\rightarrow$ But tolerable

3.6

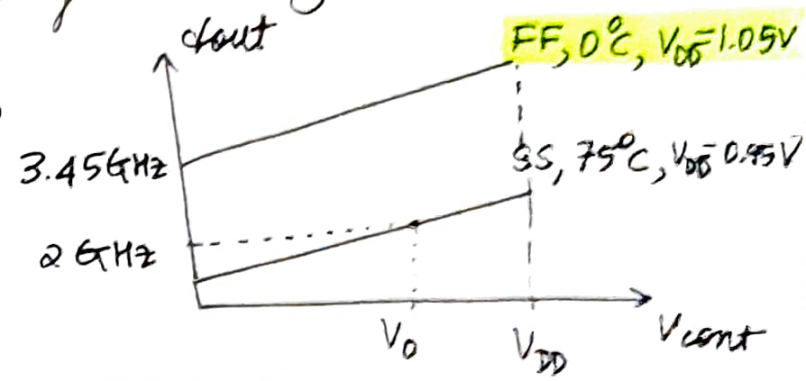
Discrete Frequency Tuning

- to wide tuning range
- Combine analog & digital tuning

add discrete f... tuning to

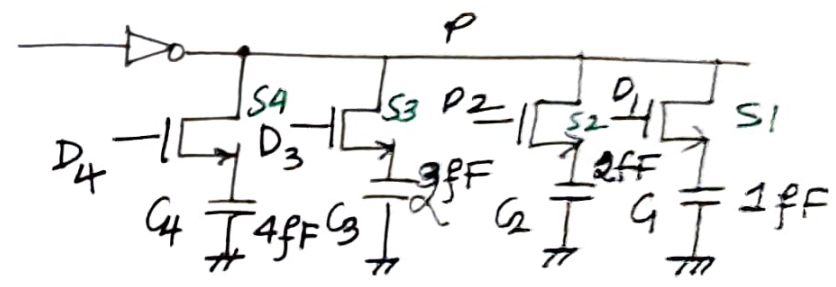


tuning range ~ 400MHz



How is it implemented ???

→ Add programmable capacitors at each node



C₁ - C₄ : programmable capacitors

* Switches driven by "thermometric" code

* As more switches are ON : C increases C₁, C₁+C₂... & decreases

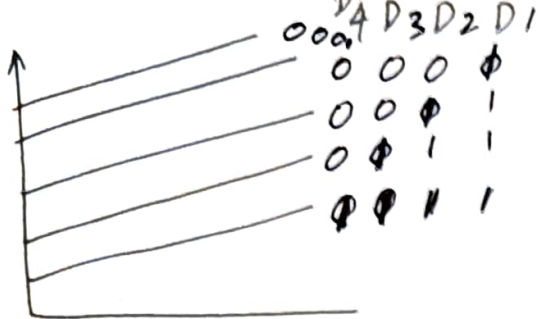
D ₄	D ₃	D ₂	D ₁	Load Cap
0	0	0	0	0
0	0	0	1	1 fF
0	0	1	1	3 fF
0	1	1	1	5 fF
1	1	1	1	9 fF

small switches width

no glitches ← Monotonic behaviour

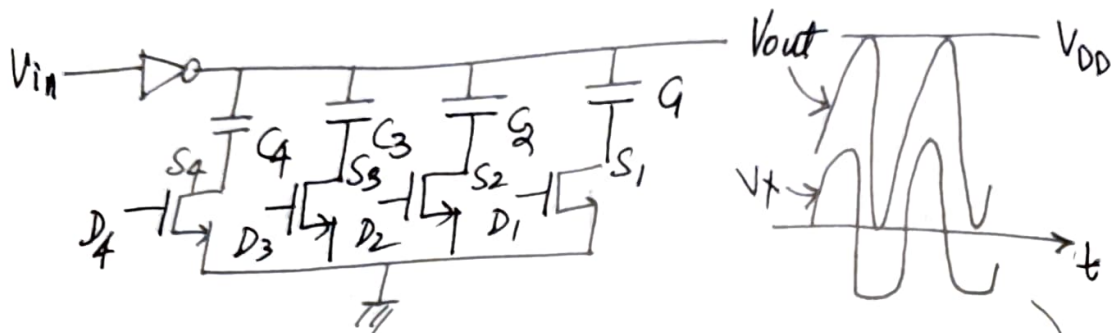
How to Select Cap Values?

- C₁ smallest ; C₄ largest
- C₃ = C₂
- 1st step gives big freq drop



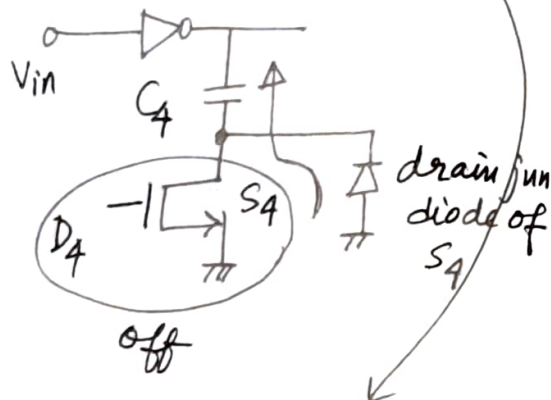
Example 3.20 Topology comparison

3.43(a)



* if switches are off

- ⊗ drain junction leakage
- ⊗ subthreshold ~~leakage~~ leakage
- ⊗ unwanted loading
- ⊗ non-zero avg voltage.



turns ON

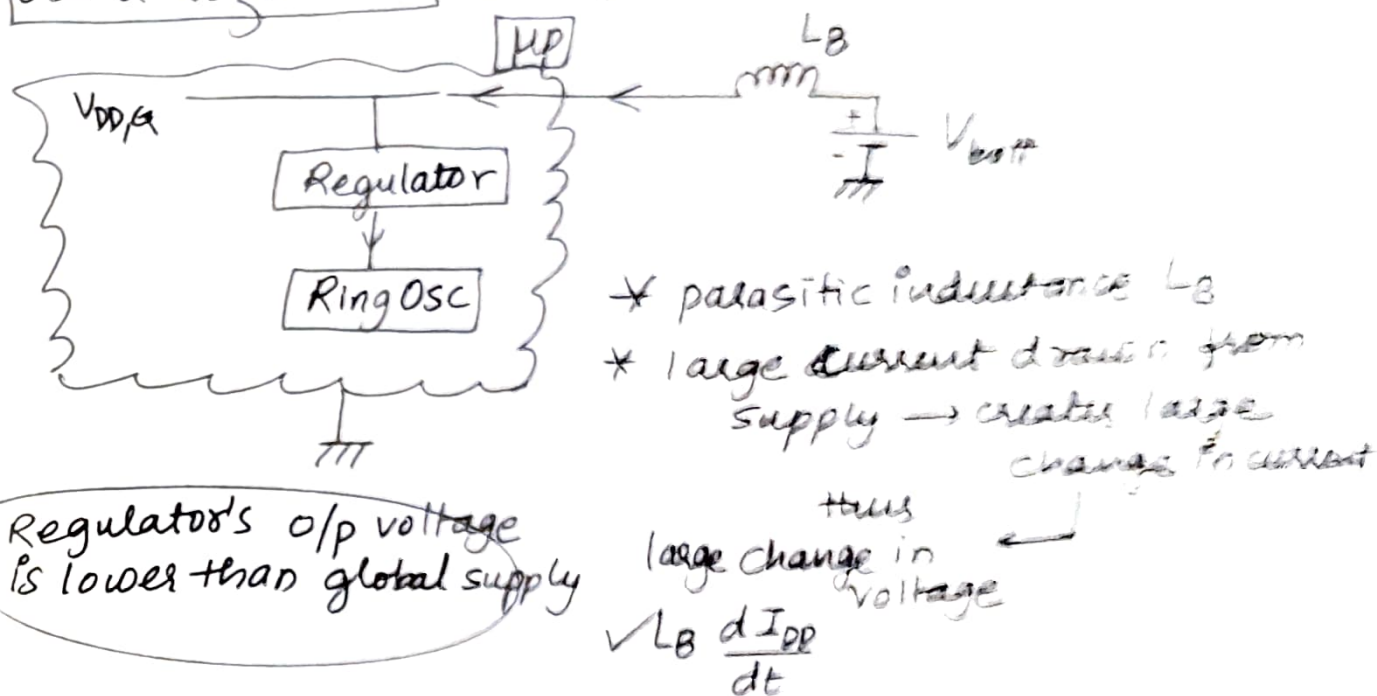
$V_{out} : 0 \text{ to } V_{DD}$

$V_x : \text{above zero to below zero}$

3.7 Problem of Supply Noise

Why? → Ring Osc sensitive to supply Voltage
 → V_{DD} noise (i.e., digital logic/micropro.)
 modulates directly freq. of Osc & phase noise

Use a Regulator to mitigate noise



3.7.1 Voltage Regulation LDO

* high-gain servo loop consists of m_1 — resistor divider — PMOS

* A_1 ensures: $V_1 \approx V_{ref}$

$$\therefore V_{DD} \approx V_{ref} \left(1 + \frac{R_1}{R_2} \right) \rightarrow \text{Low-Dropout despite change in } V_{DD,G} \text{ } V_{DD} \text{ remains regulated}$$

* V_{ref} : can be generated from Bandgap circuit

* A : 2-stage op amp

NOTE

- choose wide pass transistor to minimise voltage drop
- op amp: higher gain for better regulation
 : less noise for better phase noise

Limitations • Regulator adds its own flicker noise
— degrades osc pN

• o/p voltage of reg is lower than global supply.

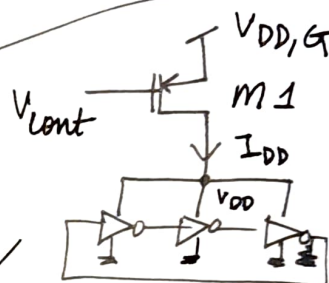
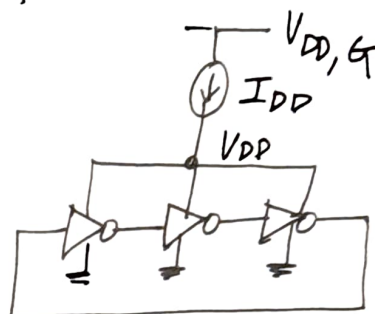
3.7.2 Current Regulation

→ isolate Ring osc from supply noise & is to feed circuit by a current source

→ I_{DD} consumes ~~some~~ some voltage headroom

$$V_{DD} = \frac{I_{DD}}{3 f_0 C_{tot}}$$

→ change to using current

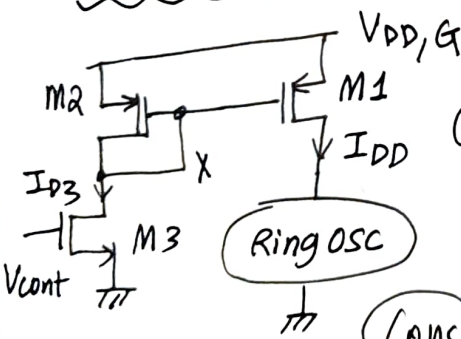


- M1 in SR
- Issues faced
 - ① noise in $V_{DD,G}$ modulates V_{gs} & I_{DD}
 - ② $1/f$ & thermal noise of $M1$

④ $V_{cont} \neq g_{round}$
 $M1 \rightarrow$ triode

③ V_{cont} cant reach $V_{DD,G} \rightarrow M1$ OFF!

Current Mirror



① Noise on $V_{DD,G}$ has negligible effect on I_{DD} if channel length modu... of $M1$ & $M3$ is SMALL

Cons

- Adds its own noise
- V_{cont} has lower & upper bounds
- Tuning Range & noise perfo. worse than LDO.